PRACTICAL -5

Aim:

To implement data storage structures using distributed storage paradigms by partitioning a large analytical dataset using Hive-style partitioning based on year and region, storing the partitioned data in MinIO object storage using Python and Boto3, querying specific partitions, and comparing the performance with a full-table scan.

Code Screenshot:

create_partitions.py

scripts > create_partitions.py

```python
import pandas as pd
import os

INPUT_FILE = "../data/metrics.csv"
OUTPUT_DIR = "../data/partitions"

df = pd.read_csv(INPUT_FILE)

print("Total records:", len(df))

for (year, month, region), group in df.groupby(
    ["year", "month", "region"]
):

    partition_path = os.path.join(
        OUTPUT_DIR,
        f"year={year}",
        f"month={month}",
        f"region={region}"
    )

    os.makedirs(partition_path, exist_ok=True)

    file_path = os.path.join(
        partition_path,
        "data.csv"
    )

    group.to_csv(file_path, index=False)

print("Partitioning completed successfully!")
```

generate_data.py X

scripts > generate_data.py > ...

```python
import pandas as pd
import random
from datetime import datetime, timedelta

NUM_RECORDS = 20000

regions = [
    "Gujarat",
    "Maharashtra",
    "Rajasthan",
    "Delhi",
    "Karnataka"
]

applications = [
    "MobileApp",
    "WebApp",
    "API",
    "Payment",
    "Analytics"
]

metrics = [
    "response_time",
    "cpu_usage",
    "memory_usage",
    "request_count",
    "error_count"
]

start_date = datetime(2022, 1, 1)

data = []

```

generate_data.py X

scripts > generate_data.py > ...

```python
data = []

for i in range(NUM_RECORDS):

    timestamp = start_date + timedelta(
        minutes=random.randint(0, 4 * 365 * 24 * 60)
    )

    data.append({
        "timestamp": timestamp,
        "year": timestamp.year,
        "month": timestamp.month,
        "region": random.choice(regions),
        "application": random.choice(applications),
        "metric": random.choice(metrics),
        "value": round(random.uniform(10, 1000), 2)
    })

df = pd.DataFrame(data)

df.to_csv("../data/metrics.csv", index=False)

print("Dataset generated successfully!")
print("Records:", len(df))
print()
print(df.head())
```

minio_query.py

scripts > minio_query.py > ...

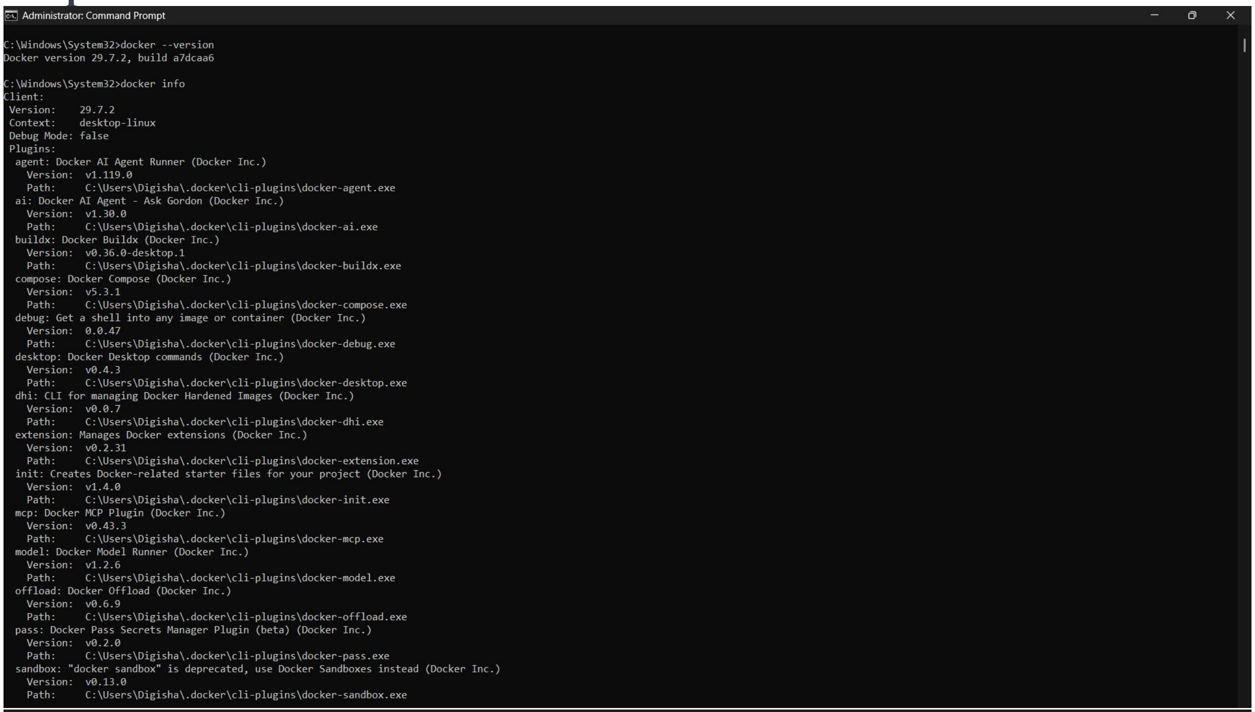
```python
import boto3
import pandas as pd
import io
import time

s3 = boto3.client(
    "s3",
    endpoint_url="http://localhost:9000",
    aws_access_key_id="admin",
    aws_secret_access_key="Minio@12345"
)

BUCKET = "metrics"

YEAR = 2025
REGION = "Gujarat"

prefix = f"year={YEAR}/"

start_time = time.perf_counter()

response = s3.list_objects_v2(
    Bucket=BUCKET,
    Prefix=prefix
)

matching_files = []

for obj in response.get("Contents", []):

    key = obj["Key"]

    if (
        f"region={REGION}" in key
```

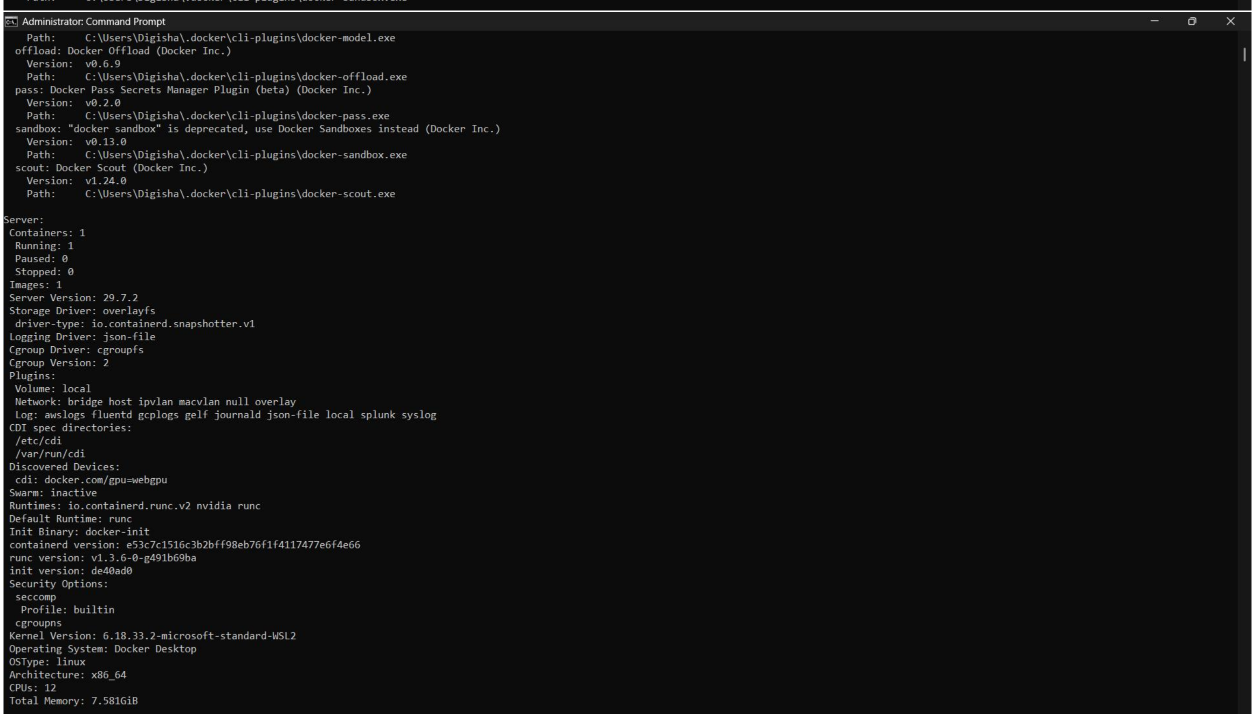
partition_query.py

scripts > partition_query.py > ...

```python
import pandas as pd
import glob
import time

YEAR = 2025
REGION = "Gujarat"

pattern = (
    f"../data/partitions/"
    f"year={YEAR}/"
    f"month=*/"
    f"region={REGION}/"
    f"data.csv"
)

files = glob.glob(pattern)

print("Partitions found:", len(files))

start_time = time.perf_counter()

dataframes = []

for file in files:
    df = pd.read_csv(file)
    dataframes.append(df)

result = pd.concat(dataframes, ignore_index=True)

end_time = time.perf_counter()

elapsed = end_time - start_time

print("Records found:", len(result))
```

Output Screenshot:





```
C:\Windows\System32>docker run -d --name minio -p 9000:9000 -p 9001:9001 -e MINIO_ROOT_USER=admin -e MINIO_ROOT_PASSWORD=Minio@12345 minio/minio server /data --console-address ":9001"

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/minio" is already in use by container "b3e2e262b460ec00befa5bfc8518422e845cc932468f242740fbebce5e8e18b3". You have to remove (or rename) that co
ntainer to be able to reuse that name.

Run 'docker run --help' for more information

C:\Windows\System32>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Windows\System32>docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED        STATUS                       PORTS                                NAMES
b3e2e262b460   minio/minio   "/usr/bin/docker-ent…"   20 hours ago   Exited (255) 4 minutes ago   0.0.0.0:9000-9001->9000-9001/tcp   minio

C:\Windows\System32>docker start minio
minio

C:\Windows\System32>docker ps
CONTAINER ID   IMAGE         COMMAND                  CREATED        STATUS         PORTS                                                      NAMES
b3e2e262b460   minio/minio   "/usr/bin/docker-ent…"   20 hours ago   Up 8 seconds   0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp   minio

C:\Windows\System32>
```

```
Windows PowerShell
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PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python --version
Python 3.14.3
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> pip --version
pip 26.1.1 from C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages\pip (python 3.14)
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> pip install boto3 pandas
Requirement already satisfied: boto3 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (1.43.69)
Requirement already satisfied: pandas in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (3.0.2)
Requirement already satisfied: botocore<1.44.0,>=1.43.69 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from boto3) (1.43.69
)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from boto3) (1.1.0)
Requirement already satisfied: s3transfer<0.20.0,>=0.19.0 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from boto3) (0.19.2
)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from botocore<1.44
.0,>=1.43.69->boto3) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from botocore<1.44.
0,>=1.43.69->boto3) (2.6.3)
Requirement already satisfied: six>=1.5 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from python-dateutil<3.0.0,>=2.1->bot
ocore<1.44.0,>=1.43.69->boto3) (1.17.0)
Requirement already satisfied: numpy>=2.3.3 in C:\Users\Digisha\AppData\Local\Programs\Python\Python314\Lib\site-packages (from pandas) (2.4.4)
Requirement already satisfied: tzdata in C:\Users\Digisha\AppData\Roaming\Python\Python314\site-packages (from pandas) (2025.3)

[notice] A new release of pip is available: 26.1.1 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python -c "import boto3; import pandas; print('Boto3 and Pandas OK')"
Boto3 and Pandas OK
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> ls


    Directory: D:\sem 5\data engineering\practical\p5\minio_partition_lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        12-08-2026     10:58                data
d-----        12-08-2026     10:58                scripts
-a----        10-08-2026     15:55            332 test_minio.py


PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python generate_data.py
Dataset generated successfully!
```
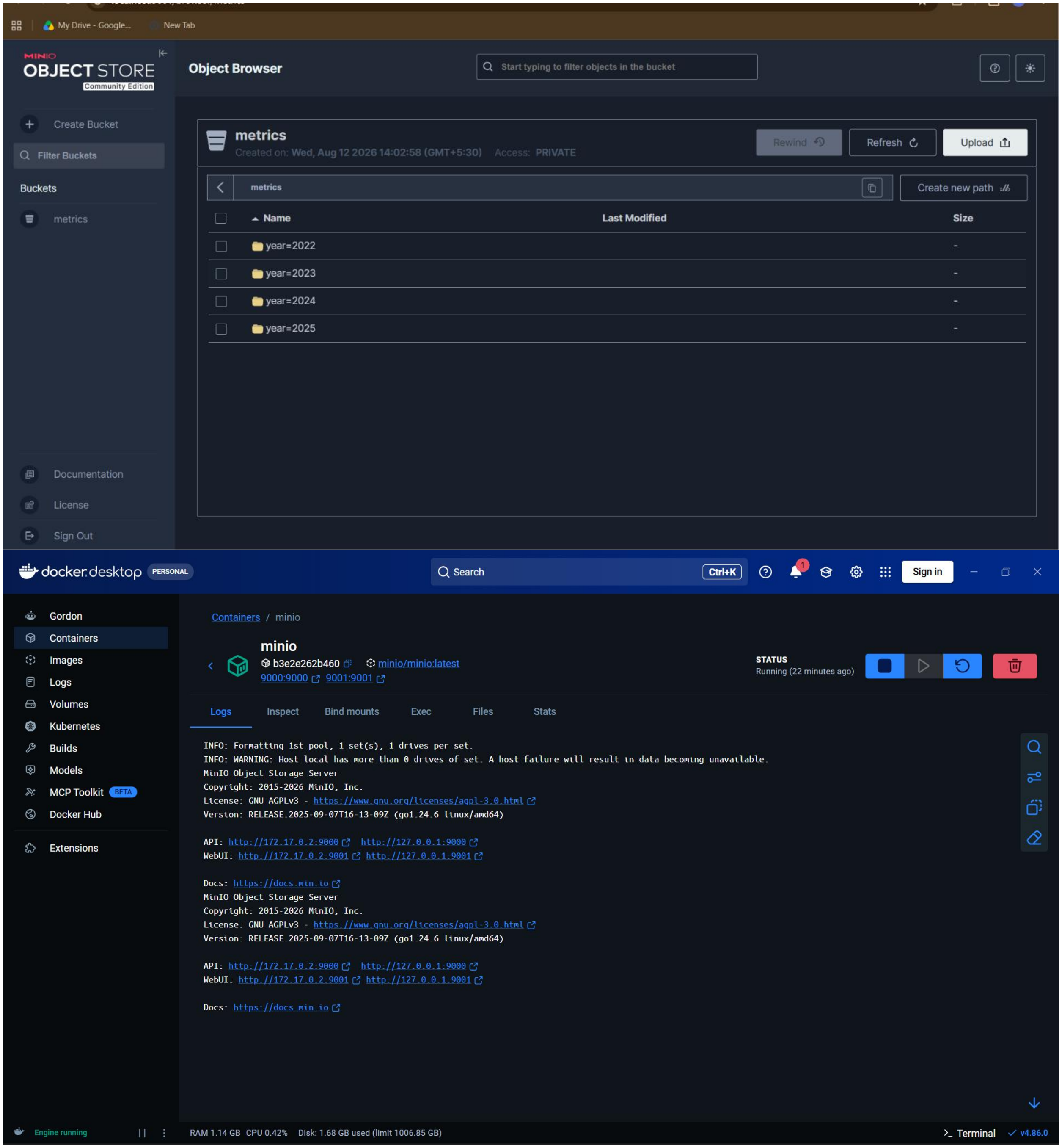

```
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python generate_data.py
Dataset generated successfully!
Records: 20000

            timestamp  year  month       region application         metric   value
0 2022-01-02 11:08:00  2022      1    Karnataka   Analytics  request_count  458.63
1 2025-09-01 07:04:00  2025      9    Rajasthan   MobileApp  response_time  636.49
2 2023-09-30 01:19:00  2023      9      Gujarat     Payment  request_count  145.37
3 2024-04-21 07:48:00  2024      4  Maharashtra   MobileApp      cpu_usage  913.09
4 2025-03-18 05:20:00  2025      3      Gujarat   Analytics      cpu_usage  596.43
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> cd ..
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python -c "import pandas as pd; df=pd.read_csv('data/metrics.csv'); print(df.shape); print(df
.head())"
(20000, 7)
             timestamp  year  month       region application         metric   value
0  2022-01-02 11:08:00  2022      1    Karnataka   Analytics  request_count  458.63
1  2025-09-01 07:04:00  2025      9    Rajasthan   MobileApp  response_time  636.49
2  2023-09-30 01:19:00  2023      9      Gujarat     Payment  request_count  145.37
3  2024-04-21 07:48:00  2024      4  Maharashtra   MobileApp      cpu_usage  913.09
4  2025-03-18 05:20:00  2025      3      Gujarat   Analytics      cpu_usage  596.43
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python partition_data.py
C:\Users\Digisha\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'D:\\sem 5\\data engineering\\practical\\p5\\minio_partition_lab\\scrip
ts\\partition_data.py': [Errno 2] No such file or directory
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> cd ..
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> python test_minio.py
Connected to MinIO successfully!
Buckets:
- metrics
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> tree data /f
Folder PATH listing for volume collage
Volume serial number is 000001EC FE20:D825
D:\SEM 5\DATA ENGINEERING\PRACTICAL\P5\MINIO_PARTITION_LAB\DATA
│   metrics.csv
│
└───partitions
    ├───year=2022
    │   ├───month=1
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
```

```
Folder PATH listing for volume collage
Volume serial number is 000001EC FE20:D825
D:\SEM 5\DATA ENGINEERING\PRACTICAL\P5\MINIO_PARTITION_LAB\DATA
│   metrics.csv
│
└───partitions
    ├───year=2022
    │   ├───month=1
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=10
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=11
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=12
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
```

```
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=5
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=6
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=7
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Karnataka
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Maharashtra
    │   │   │       data.csv
    │   │   │
    │   │   └───region=Rajasthan
    │   │           data.csv
    │   │
    │   ├───month=8
    │   │   ├───region=Delhi
    │   │   │       data.csv
    │   │   │
    │   │   ├───region=Gujarat
```

```
Windows PowerShell

PS D:\sem 5\data engineering\practical\p5\minio_partition_lab> cd scripts
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python create_partitions.py
Total records: 20000
Partitioning completed successfully!
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python upload_partitions.py
Uploading: year=2022/month=1/region=Delhi/data.csv
Uploading: year=2022/month=1/region=Gujarat/data.csv
Uploading: year=2022/month=1/region=Karnataka/data.csv
Uploading: year=2022/month=1/region=Maharashtra/data.csv
Uploading: year=2022/month=1/region=Rajasthan/data.csv
Uploading: year=2022/month=10/region=Delhi/data.csv
Uploading: year=2022/month=10/region=Gujarat/data.csv
Uploading: year=2022/month=10/region=Karnataka/data.csv
Uploading: year=2022/month=10/region=Maharashtra/data.csv
Uploading: year=2022/month=10/region=Rajasthan/data.csv
Uploading: year=2022/month=11/region=Delhi/data.csv
Uploading: year=2022/month=11/region=Gujarat/data.csv
Uploading: year=2022/month=11/region=Karnataka/data.csv
Uploading: year=2022/month=11/region=Maharashtra/data.csv
Uploading: year=2022/month=11/region=Rajasthan/data.csv
Uploading: year=2022/month=12/region=Delhi/data.csv
Uploading: year=2022/month=12/region=Gujarat/data.csv
Uploading: year=2022/month=12/region=Karnataka/data.csv
Uploading: year=2022/month=12/region=Maharashtra/data.csv
Uploading: year=2022/month=12/region=Rajasthan/data.csv
Uploading: year=2022/month=2/region=Delhi/data.csv
Uploading: year=2022/month=2/region=Gujarat/data.csv
Uploading: year=2022/month=2/region=Karnataka/data.csv
Uploading: year=2022/month=2/region=Maharashtra/data.csv
Uploading: year=2022/month=2/region=Rajasthan/data.csv
Uploading: year=2022/month=3/region=Delhi/data.csv
Uploading: year=2022/month=3/region=Gujarat/data.csv
Uploading: year=2022/month=3/region=Karnataka/data.csv
Uploading: year=2022/month=3/region=Maharashtra/data.csv
Uploading: year=2022/month=3/region=Rajasthan/data.csv
Uploading: year=2022/month=4/region=Delhi/data.csv
Uploading: year=2022/month=4/region=Gujarat/data.csv
Uploading: year=2022/month=4/region=Karnataka/data.csv
Uploading: year=2022/month=4/region=Maharashtra/data.csv
Uploading: year=2022/month=4/region=Rajasthan/data.csv
Uploading: year=2022/month=5/region=Delhi/data.csv
Uploading: year=2022/month=5/region=Gujarat/data.csv
Uploading: year=2022/month=5/region=Karnataka/data.csv
```

```
Windows PowerShell

Uploading: year=2025/month=7/region=Delhi/data.csv
Uploading: year=2025/month=7/region=Gujarat/data.csv
Uploading: year=2025/month=7/region=Karnataka/data.csv
Uploading: year=2025/month=7/region=Maharashtra/data.csv
Uploading: year=2025/month=7/region=Rajasthan/data.csv
Uploading: year=2025/month=8/region=Delhi/data.csv
Uploading: year=2025/month=8/region=Gujarat/data.csv
Uploading: year=2025/month=8/region=Karnataka/data.csv
Uploading: year=2025/month=8/region=Maharashtra/data.csv
Uploading: year=2025/month=8/region=Rajasthan/data.csv
Uploading: year=2025/month=9/region=Delhi/data.csv
Uploading: year=2025/month=9/region=Gujarat/data.csv
Uploading: year=2025/month=9/region=Karnataka/data.csv
Uploading: year=2025/month=9/region=Maharashtra/data.csv
Uploading: year=2025/month=9/region=Rajasthan/data.csv

All partitions uploaded successfully!
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python partition_query.py
Partitions found: 12
Records found: 1008
Partitions scanned: 12
Time: 59.45 ms
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts> python performance_test.py

========== FULL TABLE SCAN ==========
Records found: 1008
Records scanned: 20000
Time: 112.779 ms

========== PARTITIONED SCAN ==========
Records found: 81
Records scanned: 81
Time: 4.375 ms

========== PERFORMANCE COMPARISON ==========
Full table scan: 112.779 ms
Partition scan: 4.375 ms
Performance improvement: 96.12 %
PS D:\sem 5\data engineering\practical\p5\minio_partition_lab\scripts>
```


My Drive - Google...
New Tab
OBJECT STORE
Community Edition
Object Browser
Start typing to filter objects in the bucket
Create Bucket
Filter Buckets
Buckets
metrics
metrics
Created on: Wed, Aug 12 2026 14:02:58 (GMT+5:30) Access: PRIVATE
Rewind
Refresh
Upload
metrics
Create new path
Name
Last Modified
Size
year=2022
year=2023
year=2024
year=2025
Documentation
License
Sign Out
docker desktop
PERSONAL
Search
Ctrl+K
Sign in
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Containers / minio
minio
b3e2e262b460
minio/minio:latest
9000:9000
9001:9001
STATUS
Running (22 minutes ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
INFO: Formatting 1st pool, 1 set(s), 1 drives per set.
INFO: WARNING: Host local has more than 0 drives of set. A host failure will result in data becoming unavailable.
MinIO Object Storage Server
Copyright: 2015-2026 MinIO, Inc.
License: GNU AGPLv3 - https://www.gnu.org/licenses/agpl-3.0.html
Version: RELEASE.2025-09-07T16-13-09Z (go1.24.6 linux/amd64)
API: http://172.17.0.2:9000 http://127.0.0.1:9000
WebUI: http://172.17.0.2:9001 http://127.0.0.1:9001
Docs: https://docs.min.io
MinIO Object Storage Server
Copyright: 2015-2026 MinIO, Inc.
License: GNU AGPLv3 - https://www.gnu.org/licenses/agpl-3.0.html
Version: RELEASE.2025-09-07T16-13-09Z (go1.24.6 linux/amd64)
API: http://172.17.0.2:9000 http://127.0.0.1:9000
WebUI: http://172.17.0.2:9001 http://127.0.0.1:9001
Docs: https://docs.min.io
Engine running
RAM 1.14 GB CPU 0.42% Disk: 1.68 GB used (limit 1006.85 GB)
Terminal
v4.86.0

Learning Outcome:

After completing this practical, I learned how to organize analytical data using Hive-style partitioning and store it in an S3-compatible MinIO object storage system. I learned to use Python and Boto3 to upload and retrieve partitioned data, query only the required partitions, and compare partition-based access with full-table scanning. I also understood how partitioning can reduce unnecessary data access and improve the efficiency of analytical queries.